Ecole Polytechnique
Fédérale de Lausanne
EPFL

Basics of GeN-Foam

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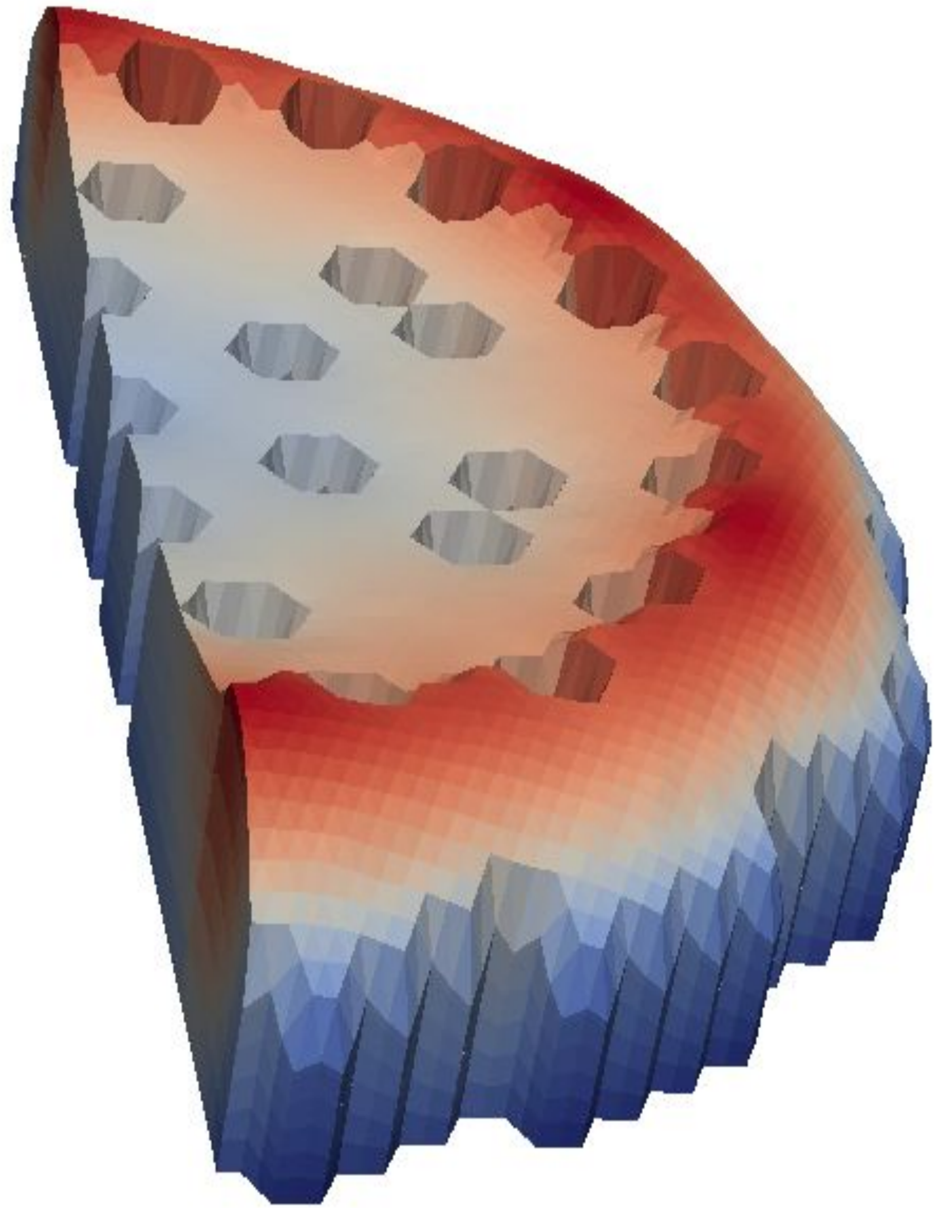
Outline

- Introduction
- Which physics?
- Multi-mesh approach
- Multi-material approach
- Coupling and coupling error
- Time stepping



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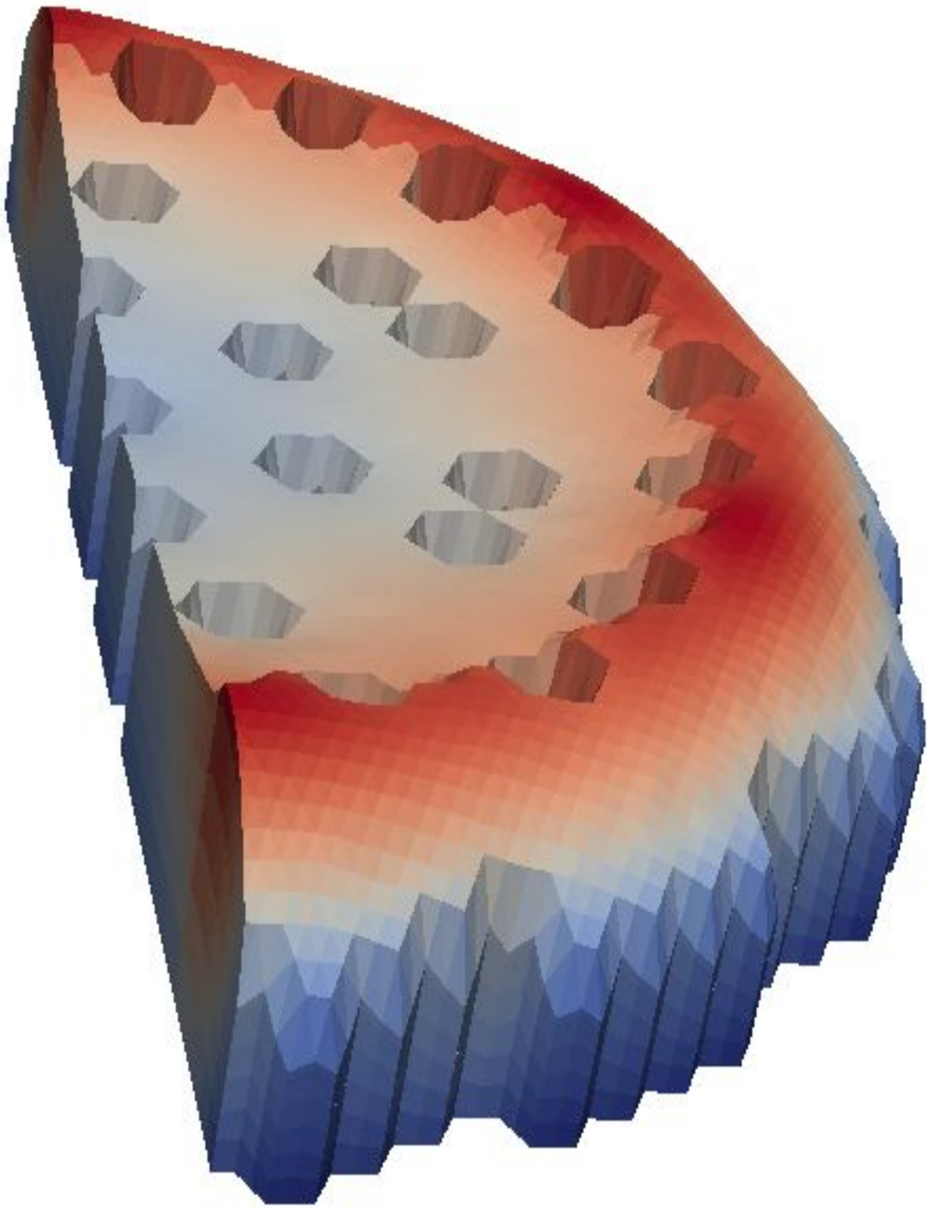
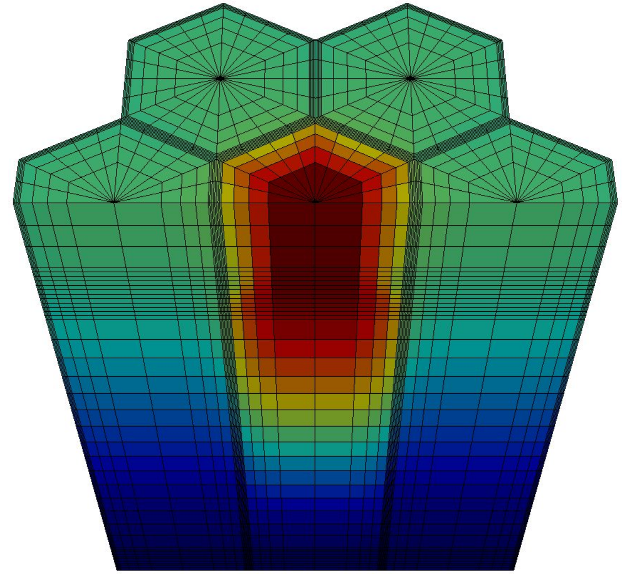
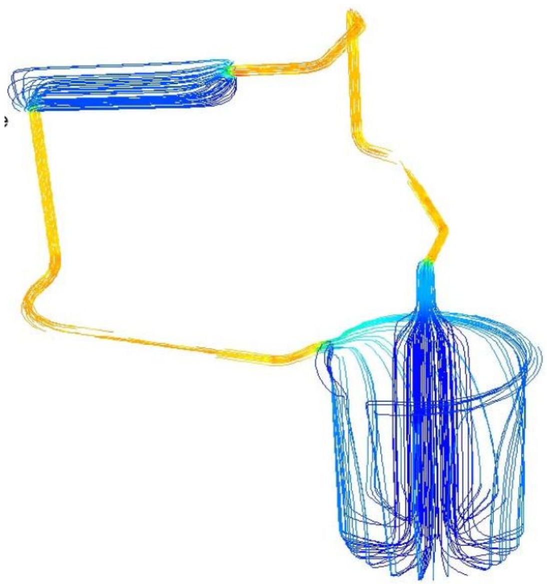
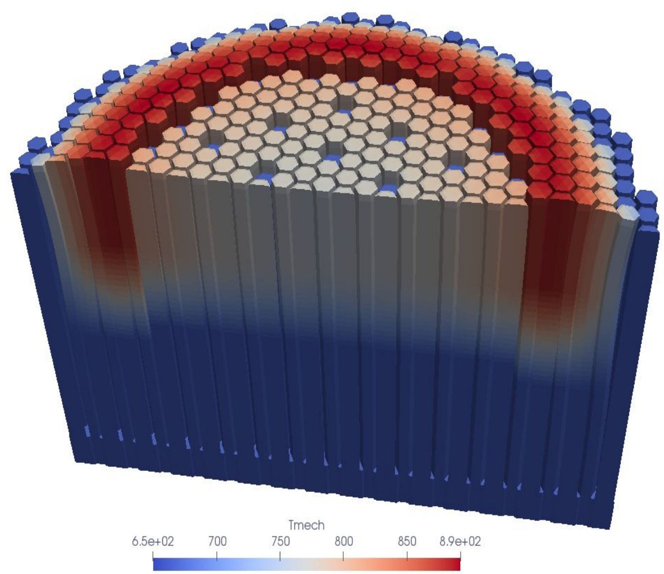


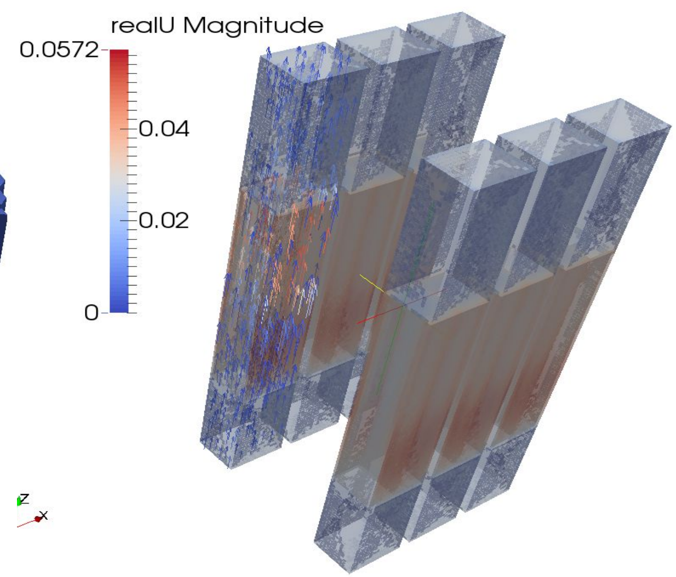
The GeN-Foam multi-physics solver:

- First attempt for a general OpenFOAM-based solver for reactor design and safety analysis
- Objective: complement legacy tool with more flexibility for novel technologies and complex situations
- Available on GitLab (foam-for-nuclear/GeN-Foam)
- Soon to be distributed also via the IAEA
- Sub-solvers as classes: can be "unplugged" and used elsewhere









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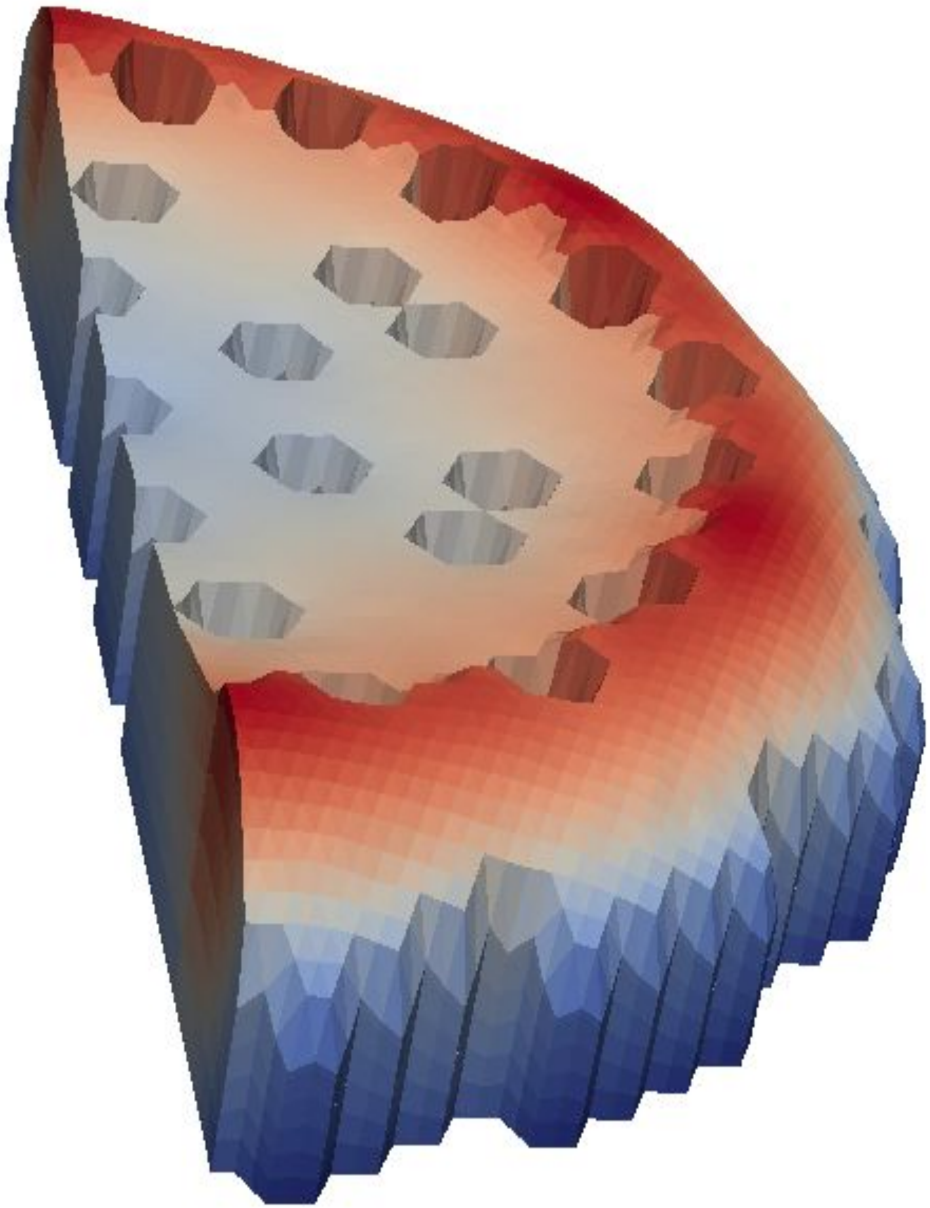
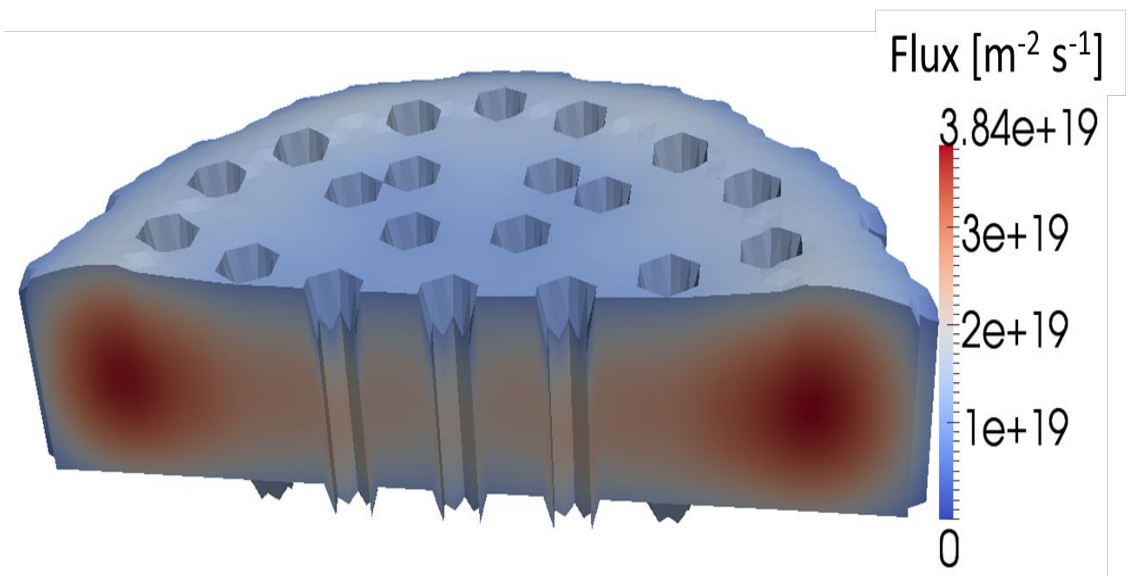
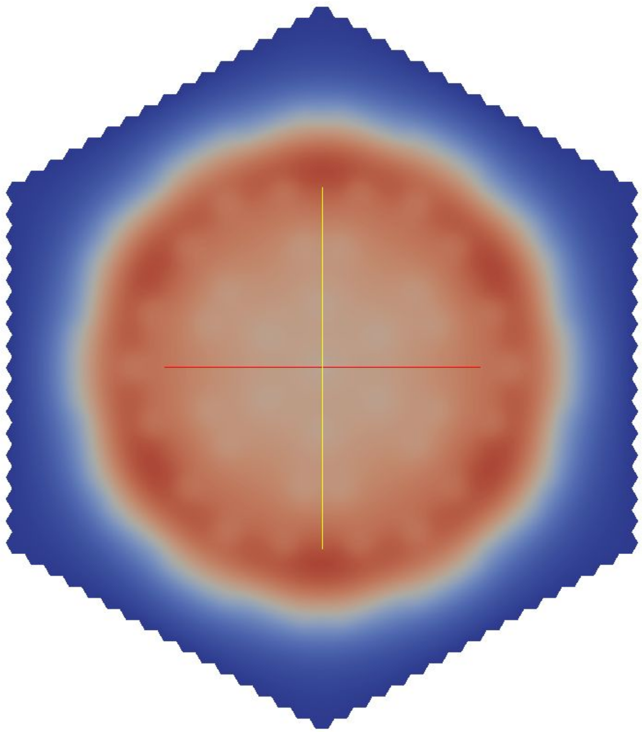


- **Which physics?**
  - ✔ Neutronics
  - ✔ Thermal-hydraulics
  - ✔ Fuel behavior
  - ✔ Core deformations (mainly for fast-reactors)

Neutronics

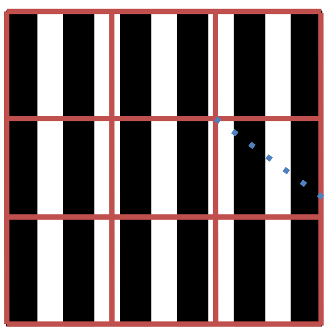
- **Point kinetics, multi-group diffusion / SP3, SN**
- **Eigenvalue** or **time dependent**
- **Parameterization** in terms of **local** temperatures, densities and deformations
- **Mesh deformed** according to displacement
- Possibility to **transport DNPs**



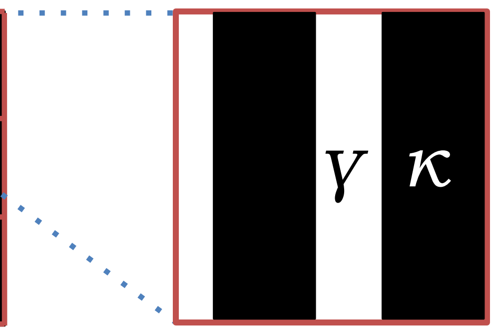


Single-phase thermal hydraulics

- **CFD:** available and ok for pools and plena, not ideal for core, HXs, etc
- Traditionally 1-D system code or sub-channel code
- Natural option for OpenFOAM: coarse-mesh porous-medium
  - ✔ 3-D version of a system code, or
  - ✔ More general version of a sub-channel code

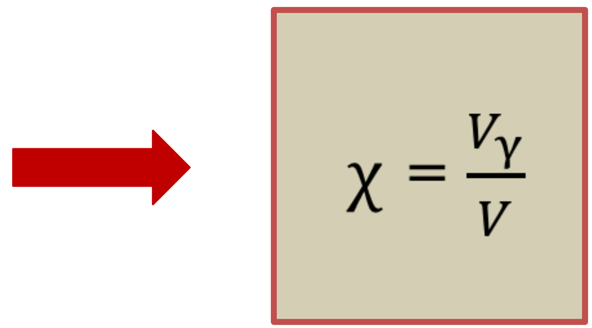


Macroscopic System



Representative Elementary Volume, V

- Fluid phase γ
- Solid phase κ (subscale structure)



$$\chi = \frac{V_\gamma}{V}$$

Porous coarse-mesh unit cell, porosity χ

What about the governing equations? And the fluid–subscale structure interaction?

Single-phase thermal hydraulics

- Obtained via volume averaging techniques:

$$\langle \cdot \rangle = \frac{1}{V} \int_{V_\gamma} \cdot \; dV$$

- Averaging locally governing RANS equations yields the macroscopic governing equations.
- By averaging derivative terms, quantities describing fluid–subscale structure interaction arise. Closure models for these terms are required.

Single-phase thermal hydraulics

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0$$

Pressure drops correlations

$$\frac{\partial(\chi\rho\boldsymbol{u})}{\partial t} + \boldsymbol{\nabla} \cdot (\chi\rho\boldsymbol{u} \otimes \boldsymbol{u}) = \boldsymbol{\nabla} \cdot (\mu_t \boldsymbol{\nabla}\boldsymbol{u}) - \boldsymbol{\nabla}(\chi p) + \chi\boldsymbol{F_g} + \chi\boldsymbol{F_{ss}}$$

Nusselt number correlations

$$\frac{\partial(\chi\rho e)}{\partial t} + \boldsymbol{\nabla} \cdot \left(\chi\rho\boldsymbol{u}\left(e + \frac{p}{\rho}\right)\right) = \boldsymbol{\nabla} \cdot (\chi k_t \boldsymbol{\nabla} T) + \boldsymbol{F_{ss}} \cdot \boldsymbol{u} + \chi\dot{Q}$$

Single-phase thermal hydraulics

$$\frac{\partial(\chi\rho\boldsymbol{u})}{\partial t} + \boldsymbol{\nabla}\cdot(\chi\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \boldsymbol{\nabla}\cdot(\mu_t\boldsymbol{\nabla}\boldsymbol{u}) - \boldsymbol{\nabla}(\chi p) + \chi\boldsymbol{F_g} + \chi\boldsymbol{F_{ss}}$$

$$\Delta p = 0.5 f_D \rho v^2 \frac{L}{D}$$

$$\frac{\Delta p}{L} = \frac{\partial p}{\partial x} = F_{ss,x} = 0.5 f_D \rho v^2 \frac{1}{D}$$

$$\boldsymbol{F_{ss}} = \boldsymbol{\kappa}(\boldsymbol{u_D})\cdot\boldsymbol{u_D}$$

$$\kappa(u_D)_{ii} = \frac{f_{D,i}\rho u_{D,i}}{2D_h\gamma^2}$$

Darcy friction factor

Single-phase thermal hydraulics

$$\frac{\partial(\chi\rho e)}{\partial t} + \boldsymbol{\nabla}\cdot\left(\chi\rho\boldsymbol{u}\left(e+\frac{p}{\rho}\right)\right) = \boldsymbol{\nabla}\cdot(\chi k_t \boldsymbol{\nabla} T) + \boldsymbol{F}_{ss}\cdot\boldsymbol{u} + \chi\dot{Q}$$

$$\dot{Q}_{ss} \propto h(T_{SS} - T)$$

Temperature of subscale structure

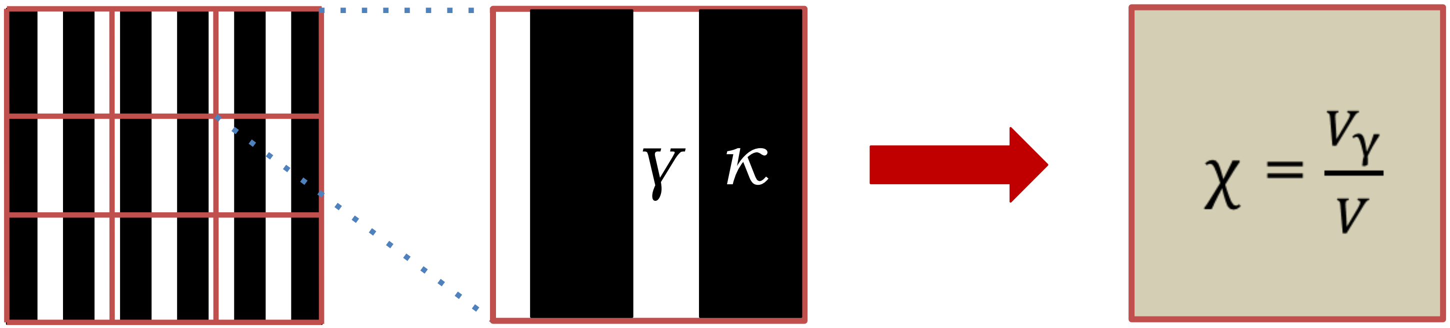
Heat transfer coefficient, h(Nu)
Nu from correlations

$h(T_{SS} - T)$ in W/m²
Multiply by volumetric area

$$\dot{Q}_{ss} = A_V h(T_{SS} - T)$$

Single-phase thermal hydraulics

- **Problem:**



- What about the structures that we have averaged?

$$\frac{\partial(\chi\rho e)}{\partial t} + \boldsymbol{\nabla}\cdot\left(\chi\rho\boldsymbol{u}\left(e+\frac{p}{\rho}\right)\right) = \boldsymbol{\nabla}\cdot(\chi k_t \boldsymbol{\nabla} T) + \boldsymbol{F_{ss}}\cdot\boldsymbol{u} + \dot{Q}$$

$$\dot{Q}_{ss} = A_V h(T_{SS} - T)$$

We need a model for Tss

Single-phase thermal hydraulics

$$\dot{Q}_{ss} = A_V h(T_{SS} - T)$$

At minimum, 0-D

$$\rho_{SS} c_{p,ss} \frac{\partial T_{SS}}{\partial t} = A_V h(T - T_{SS})$$

Or 0-D with (coarse-mesh) thermal diffusivity

$$\rho_{SS} c_{p,ss} \frac{\partial T_{SS}}{\partial t} = \nabla \cdot (\gamma \boldsymbol{k}_{\boldsymbol{SS}} \nabla T) + A_V h(T - T_{SS})$$

But not always
enough…

Single-phase thermal hydraulics

$$\dot{Q}_{ss} = A_V h(T_{SS} - T)$$

We need sub-scale models, viz.

- 1-D model for pin- and plate type fuel available in GeN-foam

Single-phase thermal hydraulics

- Same set of equations for fine and coarse mesh treatment ("porous terms" disappear in clear fluid)
  - implicit "natural" coupling between porous and clear zones

$$\frac{\partial \gamma \rho}{\partial t} + \nabla \cdot (\gamma \rho \boldsymbol{u}) = 0$$

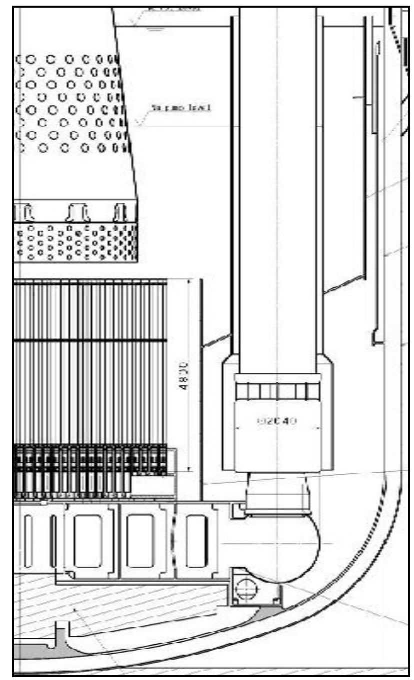
$$\frac{\partial \gamma \rho \boldsymbol{u}}{\partial t} + \nabla \cdot (\gamma \rho \boldsymbol{u} \otimes \boldsymbol{u}) = \nabla \cdot (\mu_T \nabla \boldsymbol{u}) - \nabla \gamma p + p_i \nabla \gamma + \gamma \boldsymbol{F_g} + \gamma \boldsymbol{F_{ss}}$$

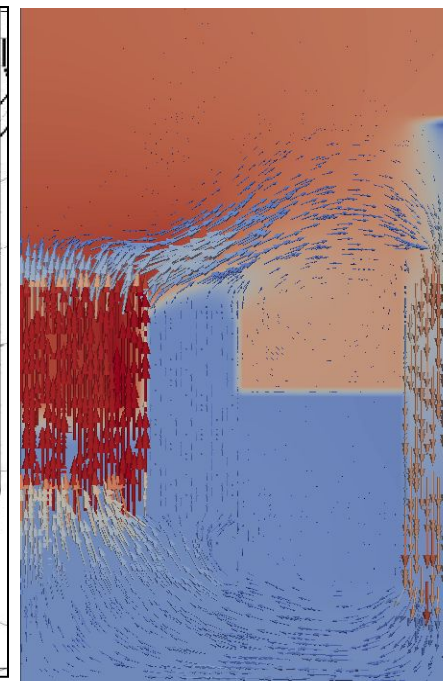
$$\boldsymbol{F_{ss}} = \boldsymbol{\kappa}(\boldsymbol{u_D}) \cdot \boldsymbol{u_D} \qquad \kappa(u_D)_{ii} = \frac{f_{D,i} \rho u_{D,i}}{2 D_h \gamma^2} \qquad f_{D,i} = A_{f_{D,i}} Re^{B_{f_{D},i}}$$

$$\frac{\partial \gamma \rho e}{\partial t} + \nabla \cdot \big(\boldsymbol{u} \gamma (\rho e + p)\big) = \nabla \cdot (\gamma k_T \nabla \mathrm{T}) + \gamma \boldsymbol{F_{ss}} \cdot \boldsymbol{u} + \gamma \dot{Q}_{ss}$$

$$\dot{Q}_{ss} = A_V h (T_{SS} - T) \qquad Nu_i = A_{Nu,i} Re^{B_{Nu,i}} Pr^{C_{Nu,i}} + D_{Nu,i}$$

$$\rho_{SS} c_{p,ss} \frac{\partial T_{SS}}{\partial t} = \nabla \cdot (\gamma \boldsymbol{k_{SS}} \nabla T) + A_V h (T - T_{SS})$$



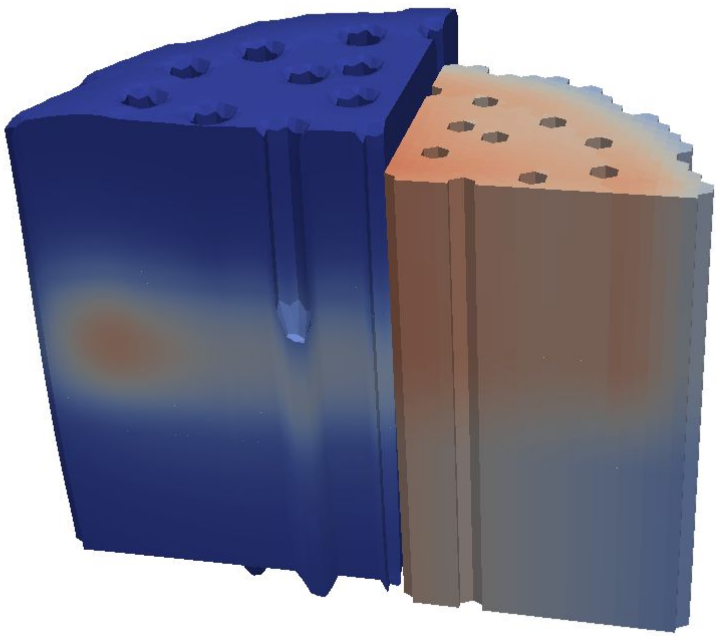


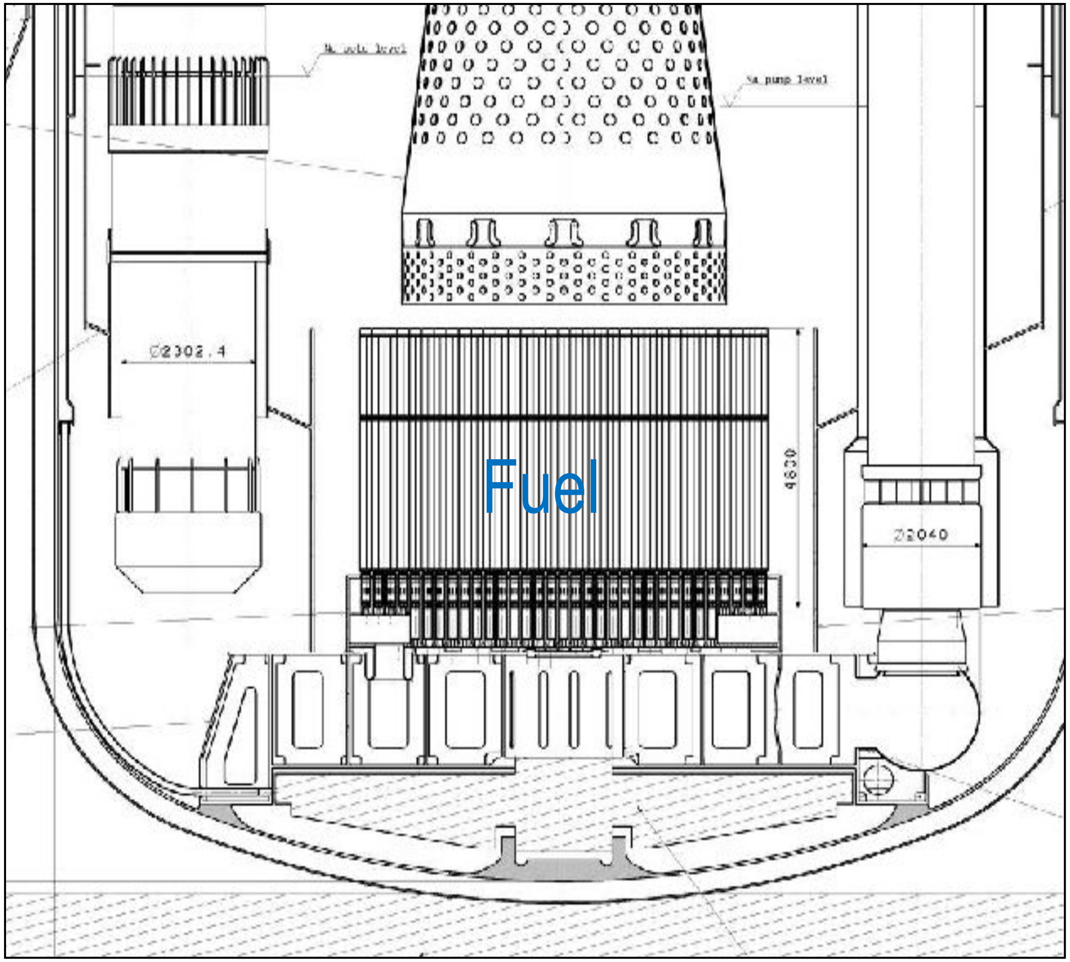
Two-phase thermal hydraulics

- Same approach as for single-phase thermal-hydraulics (porous-medium with sub-scale structure)
- Beyond the scope of this presentation. Further info soon to be published as PhD thesis of EPFL (work of Stefan Radman)

Mesh deformation

- **To displace the mesh** in fast reactor calculations

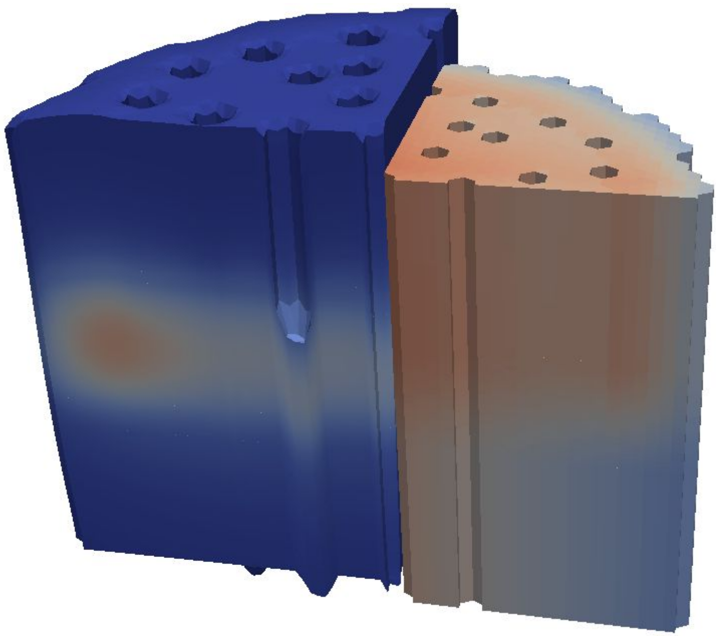


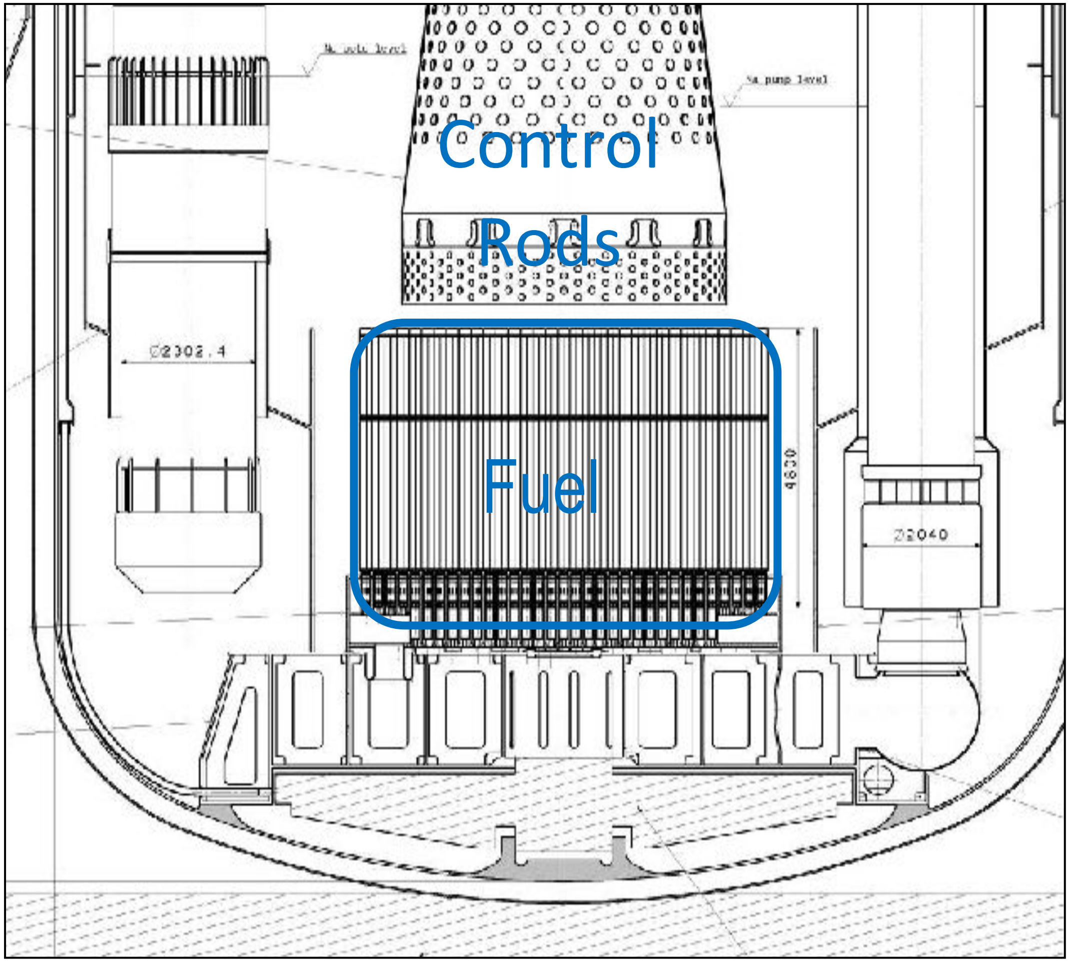


Mesh deformation

- **To displace the mesh** in fast reactor calculations
- Fuel and CR driveline expansion based on

$$v_f \cdot \nabla D_f = \alpha_{f/c}\left(T_{f/c} - T_{f/c,ref}\right)$$





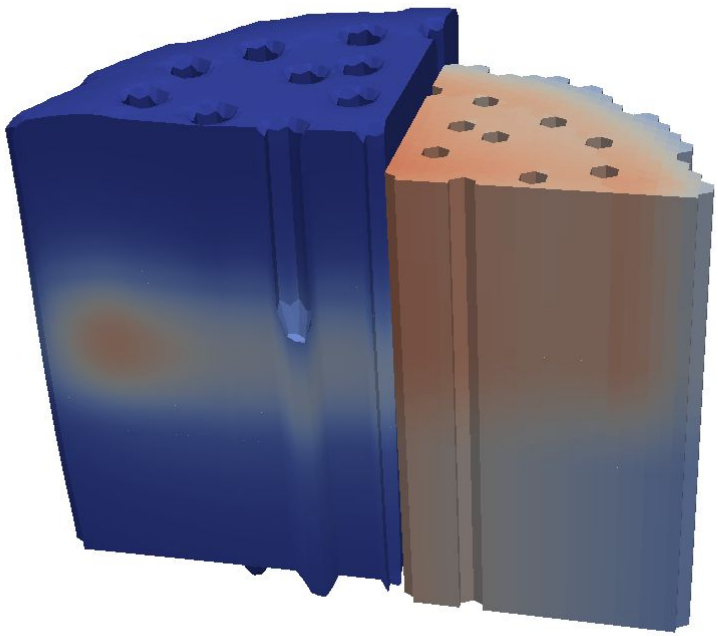
Mesh deformation

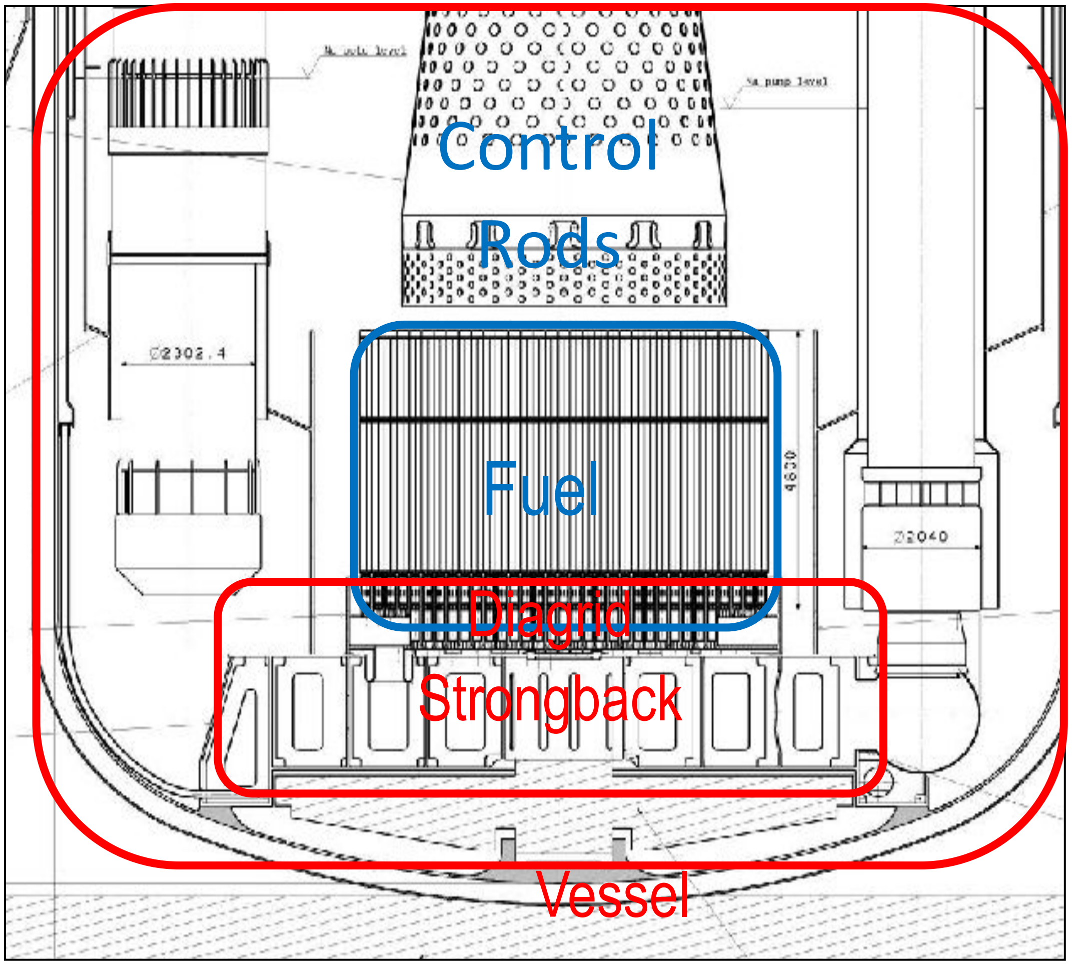
- **To displace the mesh** in fast reactor calculations
- Fuel and CR driveline expansion based on

$$v_f \cdot \nabla D_f = \alpha_{f/c}\left(T_{f/c} - T_{f/c,ref}\right)$$

- Thermo-elastic solver

for other structures





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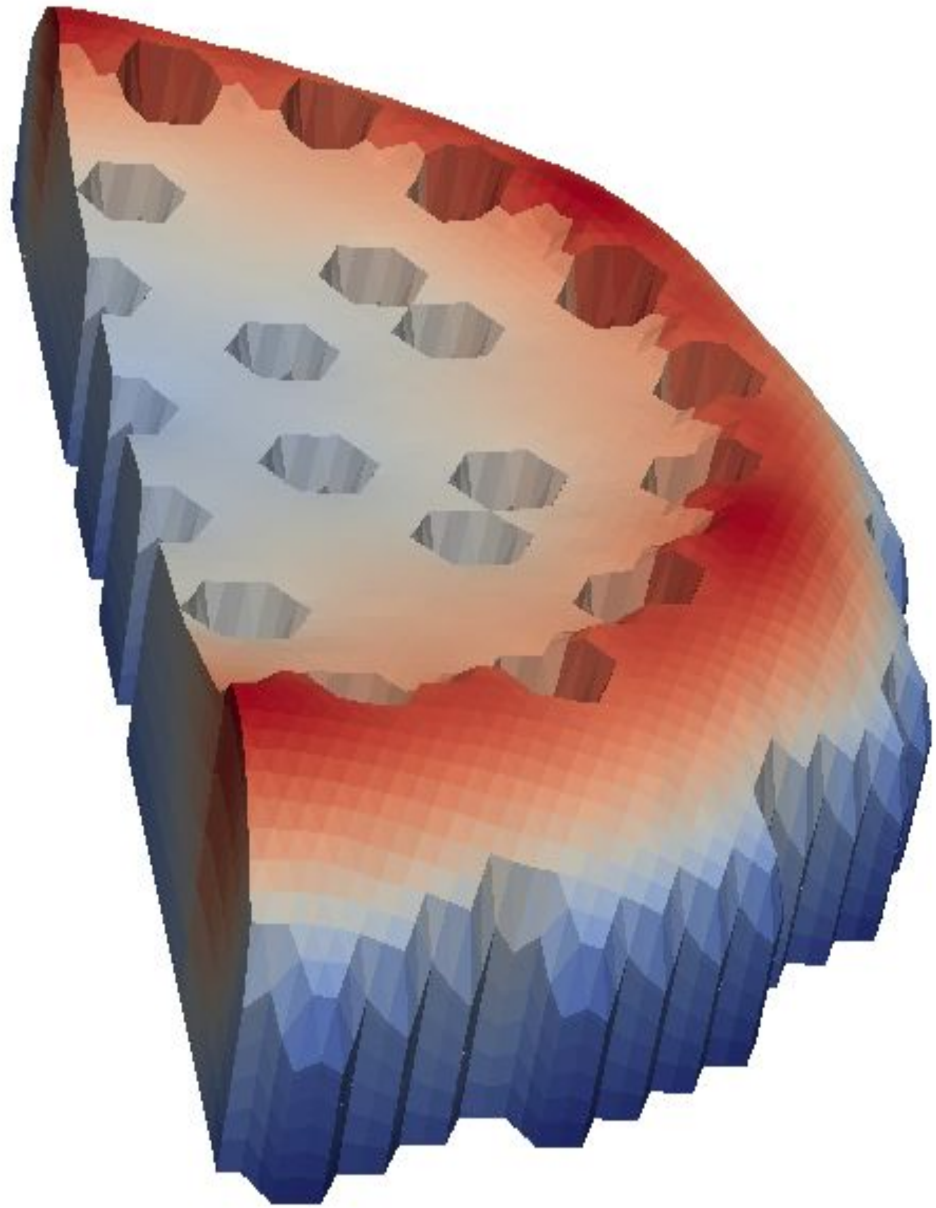
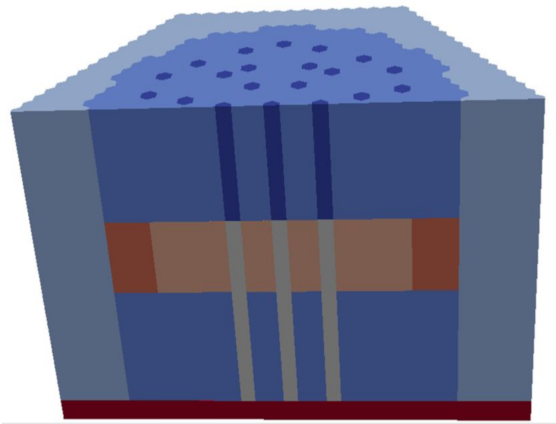
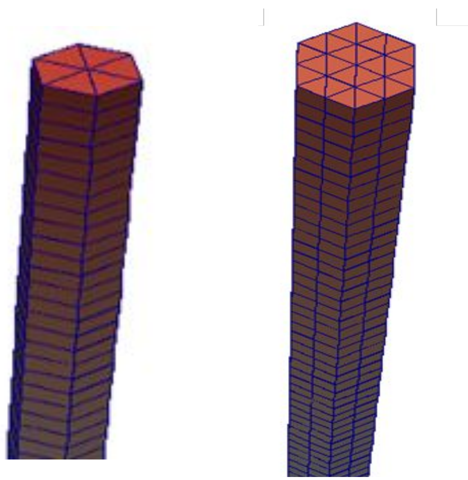
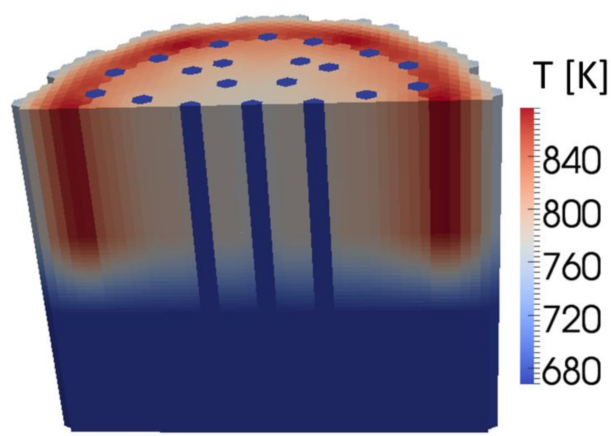


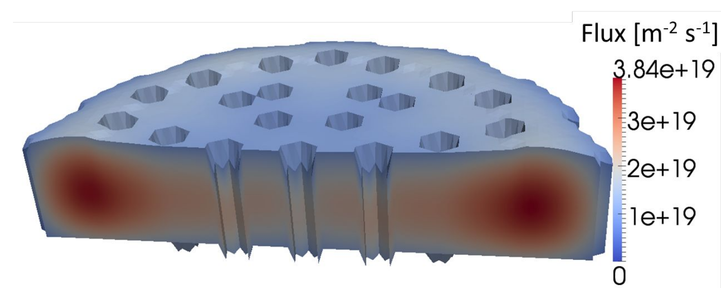
Multi-mesh



- Problem: different meshes for different “physics”
- Solution: multi-mesh (multi-region approach)
- One mesh for each “physics”
- Projection of fields from one mesh to the other for coupling (e.g., TcoolNeutro, DNeutro)







Multi-mesh: in practice

- **For the mesh:**
  - ✔ Three different meshes for neutronics, thermal-hydraulics, thermal-mechanics (with their own cellZones)

- **How is it coded:**
  - ✔ Create three different meshes
  - ✔ Associate fields to the appropriate mesh (e.g., fluxes to the neutronics mesh)
  - ✔ Create extra fields for coupling (e.g., TcoolNeutro)
  - ✔ Project fields from one mesh to the other at every coupling iteration, e.g., Tcool (thermal-hydraulic mesh) to TcooNeutro (neutronics-mesh), with careful choice of default values when no correspondence

Multi-mesh: in practice

- **Case folder:**

- Case
  - └ 0
    - └ U
    - └ T
    - └ ...
  - └ constant
    - └ turbulenceProperties
    - └ ...
  - └ system
    - └ fvSolution
    - └ fvSchemes
    - └ ...

- Case
  - └ 0
    - └ neutroRegion
      - └ Flux
      - └ ...
    - └ fluidRegion
      - └ U
      - └ ...
    - └ thermoMechanicalRegion
      - └ ...
  - └ constant
    - └ neutroRegion
    - └ fluidRegion
    - └ thermoMechanicalRegion
    - └ ...
  - └ system
    - └ neutroRegion
    - └ fluidRegion
    - └ thermoMechanicalRegion
    - └ ...

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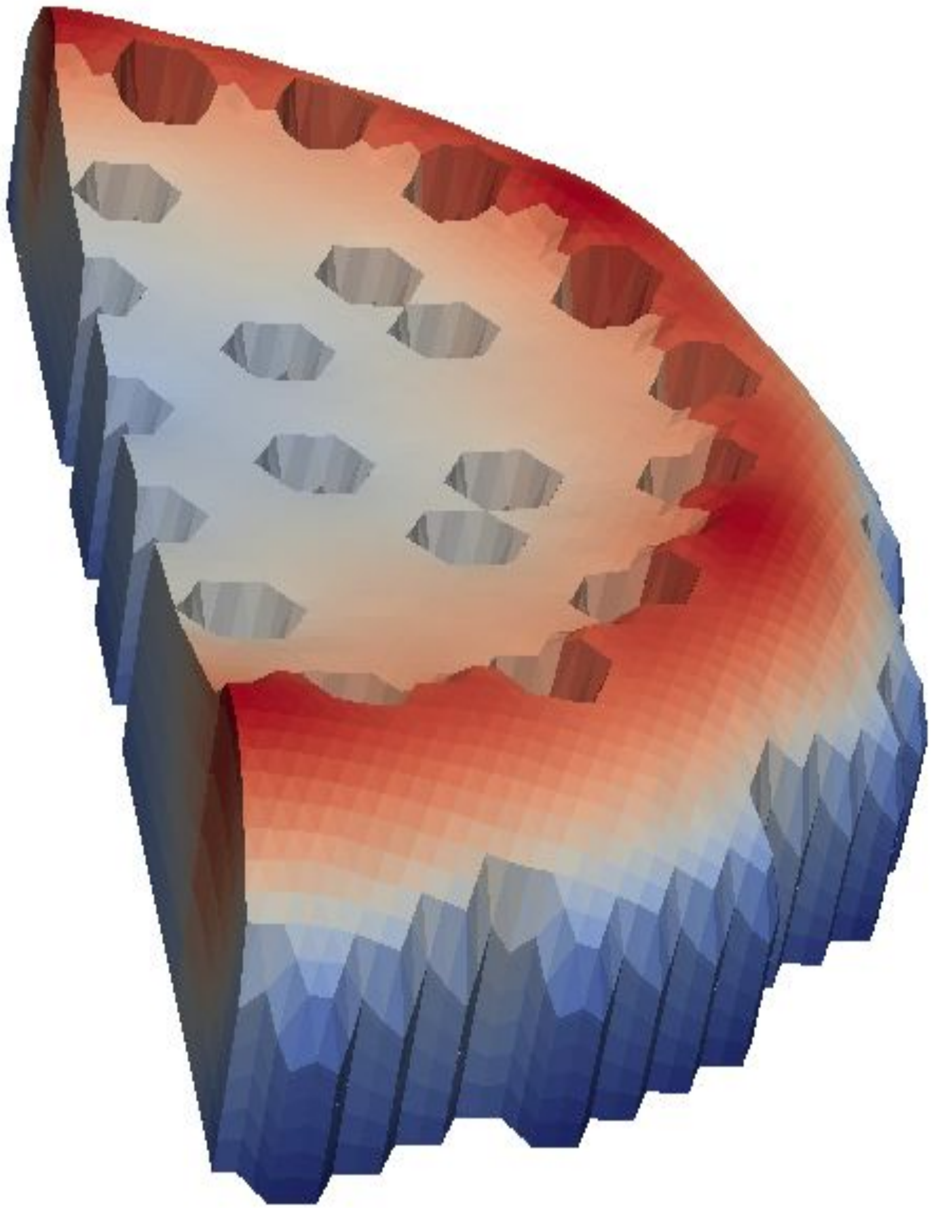
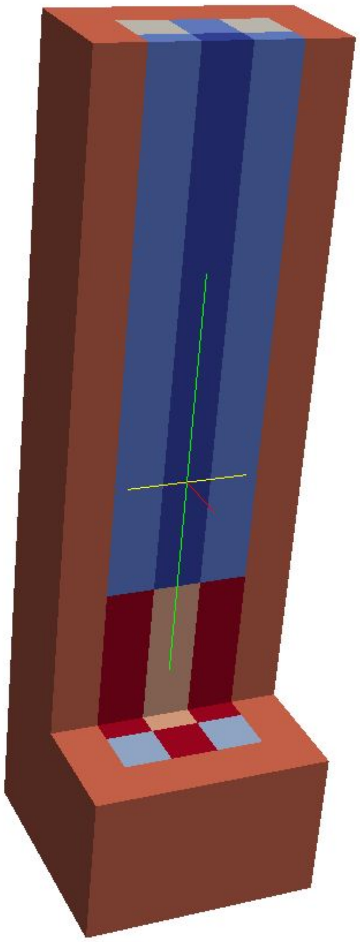


Multi-material

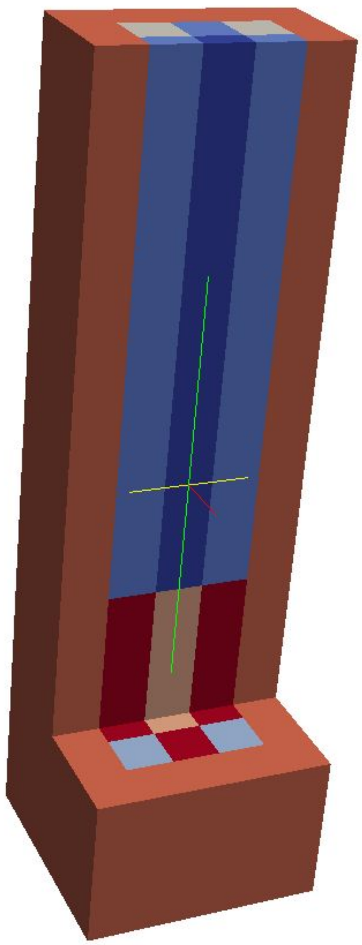


- Problem: one mesh, multiple material
- Solutions: cellZones
  - associate a label to each cell in polymesh/cellZones

```
FoamFile
{
    version     2.0;
    format      ascii;
    class       regIOobject;
    location    "constant/fluid/polyMesh";
    object      cellZones;
}
// * * * * * * * * * * * * * * * * * * * * * * *

7
(
controlRod
{
    type cellZone;
cellLabels      List<label>
5994
(
0
1
2
```

Multi-material



- Then, for each physics, an input file (dictionary) is used that associates each of these labels to a set of properties. For instance in /constant/neutroRegion/nuclearData

```
zones
 (

controlRod
{
 fuelFraction 1.000000e+00 ;
 IV nonuniform List<scalar> 1 (8.477550e-07  );
 D nonuniform List<scalar> 1 (1.562700e-02  );
 nuSigmaEff nonuniform List<scalar> 1 (0.000000e+00  );
 sigmaPow nonuniform List<scalar> 1 (0.000000e+00  );
 scatteringMatrix  1  1 (
 ( 2.509070e+01 )
 );
```

Multi-material: in practice

- **How to create a multi-zone mesh:**
  - ✔ All mesh generators allows for the option to generate "cellZones"
  - ✔ NB: cellZones are called in different ways (physical volumes in gmsh, groups in Salome, etc)
  - ✔ The mesh conversion tool (e.g., gmshTofoam) takes care of converting the format
- **How is it coded:**
  - ✔ Make a loop over cells
  - ✔ Read their label
  - ✔ Assign a value
- **Case folder:**
  - ✔ Dictionaries that associates a cellZone to some value of a field or property

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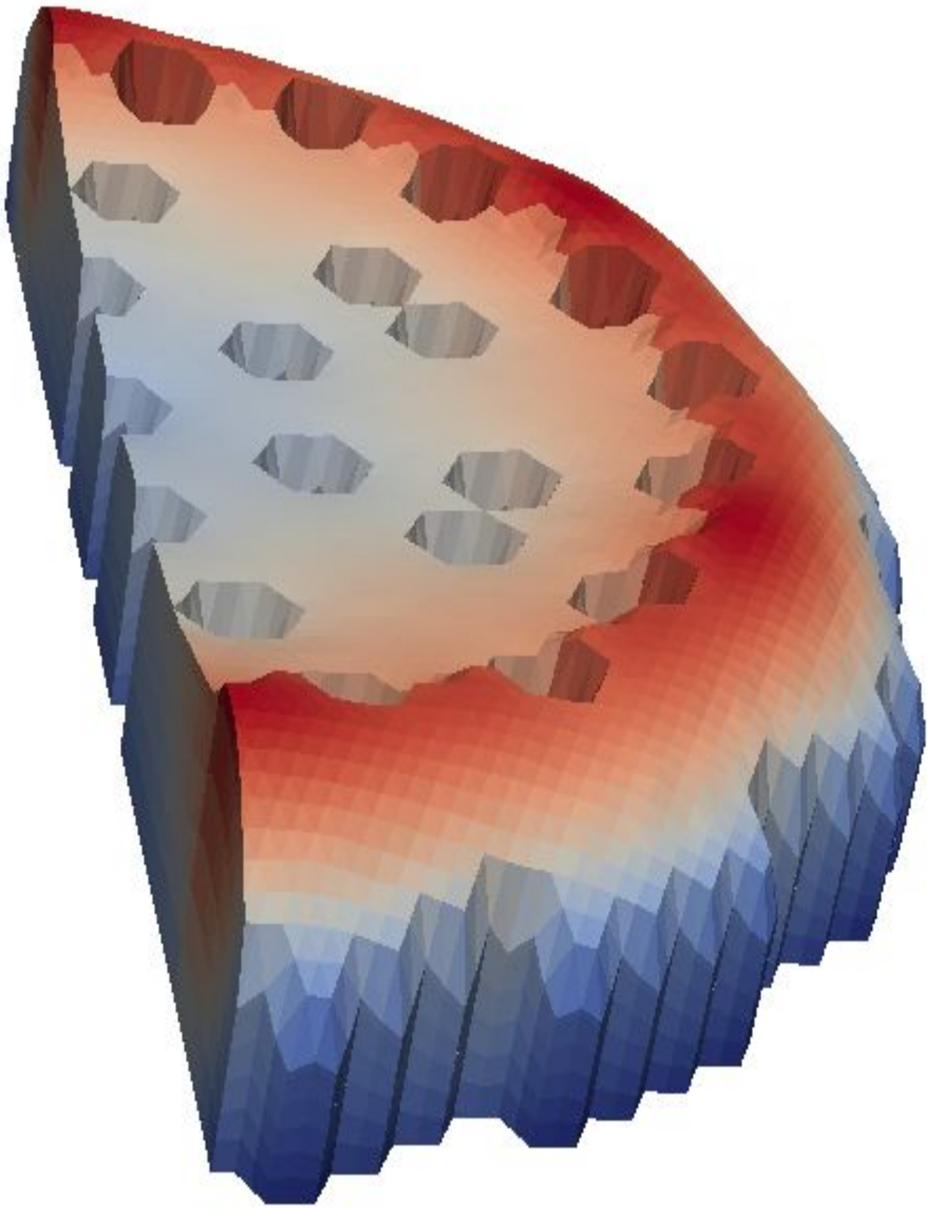


Coupling and coupling error

- **Two main options**
  - ✔ **Matrix-coupled solution:**
    - ✔ All equations in the same matrix
    - ✔ Straightforward, no iterations needed, can be faster
    - ✔ Often difficult to precondition
  - ✔ **Segregated solution (operator splitting):**
    - ✔ One matrix for each equation
    - ✔ Easier preconditioning and optimal choice of solution method
    - ✔ No need to solve all physics at each coupling/time step
  - ✔ **Possibility to combine the two**
    - ✔ e.g., segregated for each mesh, coupled for some physics (multi-group diffusion)

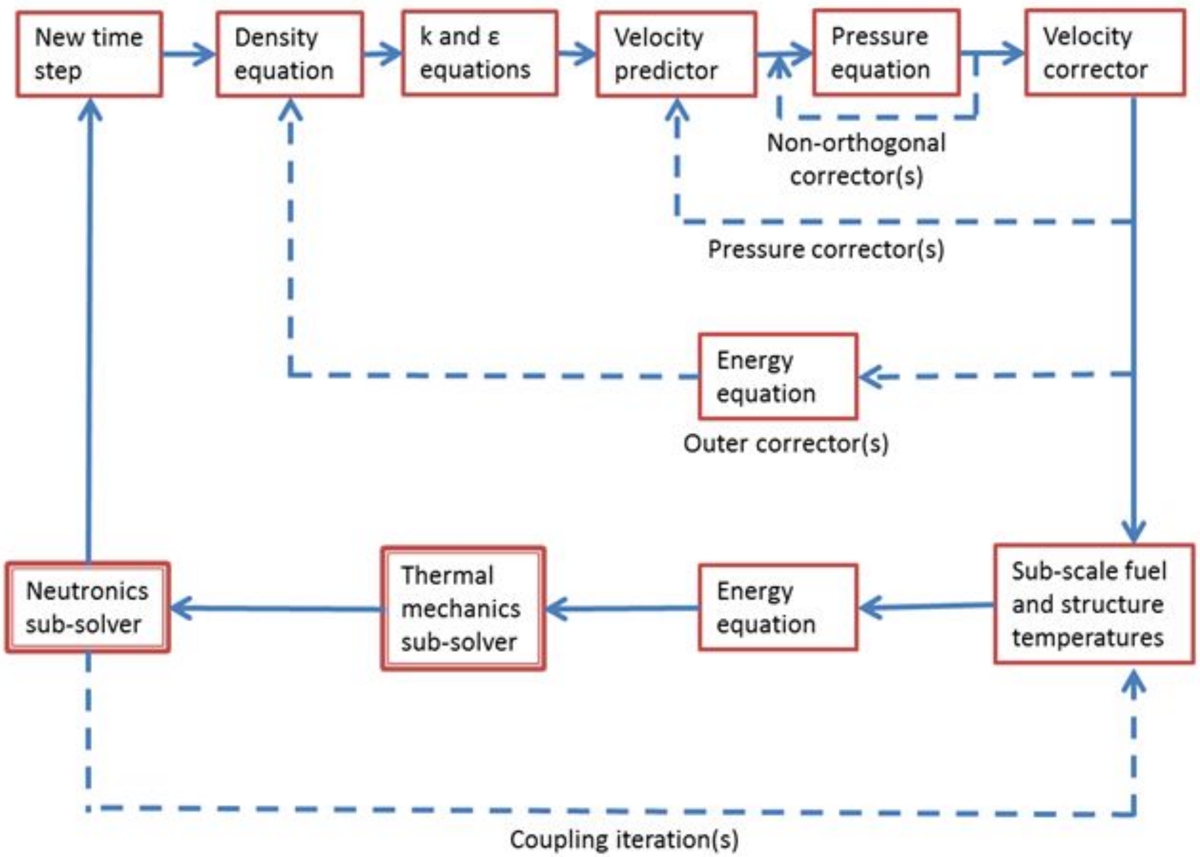
Coupling and coupling error

- Coupled solution not (fully) available in the ESI and Foundation versions of OpenFOAM

- Segregated solution employed for GeN-Foam

Coupling and coupling error: coupling strategy (single phase)

- Initial residuals evaluated before solving each iteration
- Solve only if initial residuals higher than desired residuals for each physics
- Keep iterating till initial residual for each physics is lower than the desired coupling residual



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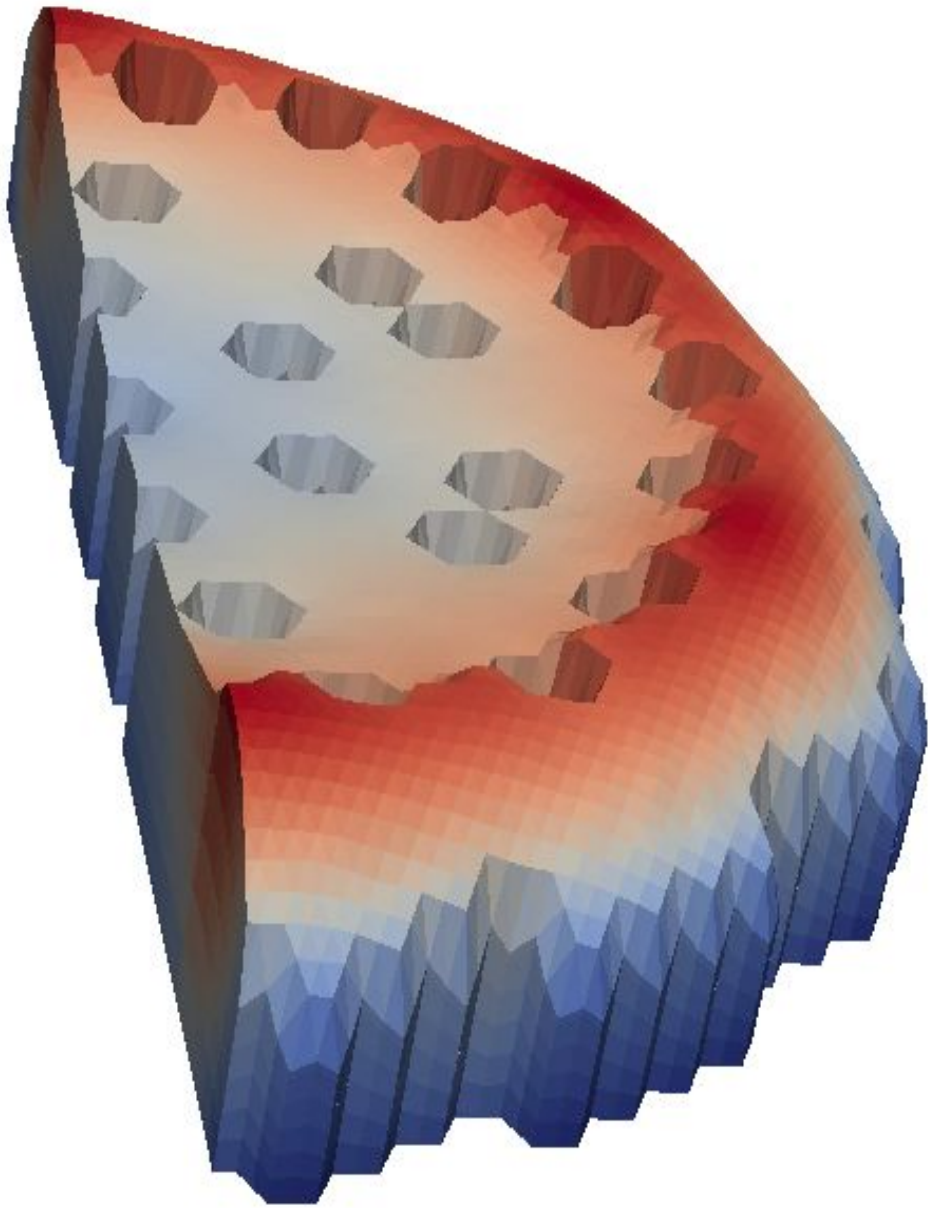

Time stepping: strategy

- **Problem: different time constants for different physics**
- **Two main options**
  - ✔ **Sub-stepping:**
    - ✔ "Slow" physics solved at every main time step
    - ✔ Multiple time steps of "fast physics" inside every "main" time step
    - ✔ Potentially faster
  - ✔ **Or not:**
    - ✔ Check on all initial residuals at every time step and decide which one to solve
    - ✔ Time stepping set by fastest physics
    - ✔ A bit slower but easier control coupling error
- **In GeN-Foam:**
  - ✔ No subs-stepping

Time stepping: choice of time step

- **Fixed or adaptive**
- **Two possibilities for adaptive time step:**
  - ✔ Based on **previous iteration**
  - ✔ Based on **current iteration,** more stable, more time consuming
- **Criteria for adaptive time step:**
  - ✔ Based on **stability criteria**: e.g. CFL condition
  - ✔ Based on **“physics based”** criteria: e.g., max power variation
  - ✔ Based on **rigorous mathematical approaches**: e.g., in high-order time discretization schemes, based on the difference between solutions for two different orders

- **In GeN-Foam:**
  - ✔ Fixed or adaptive; adaptive based on CFL + max power variation, using values at previous time step; necessary to guess first one

Thank you for your attention

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